上海交通大学
SHANGHAI JIAO TONG UNIVERSITY

Carbon Accounting

Lecture 12

Financial Carbon Accounting

Haishan Yu

饮 水 思 源 · 爱 国 荣 校



Why would a bank care about CO_2?



Scenario

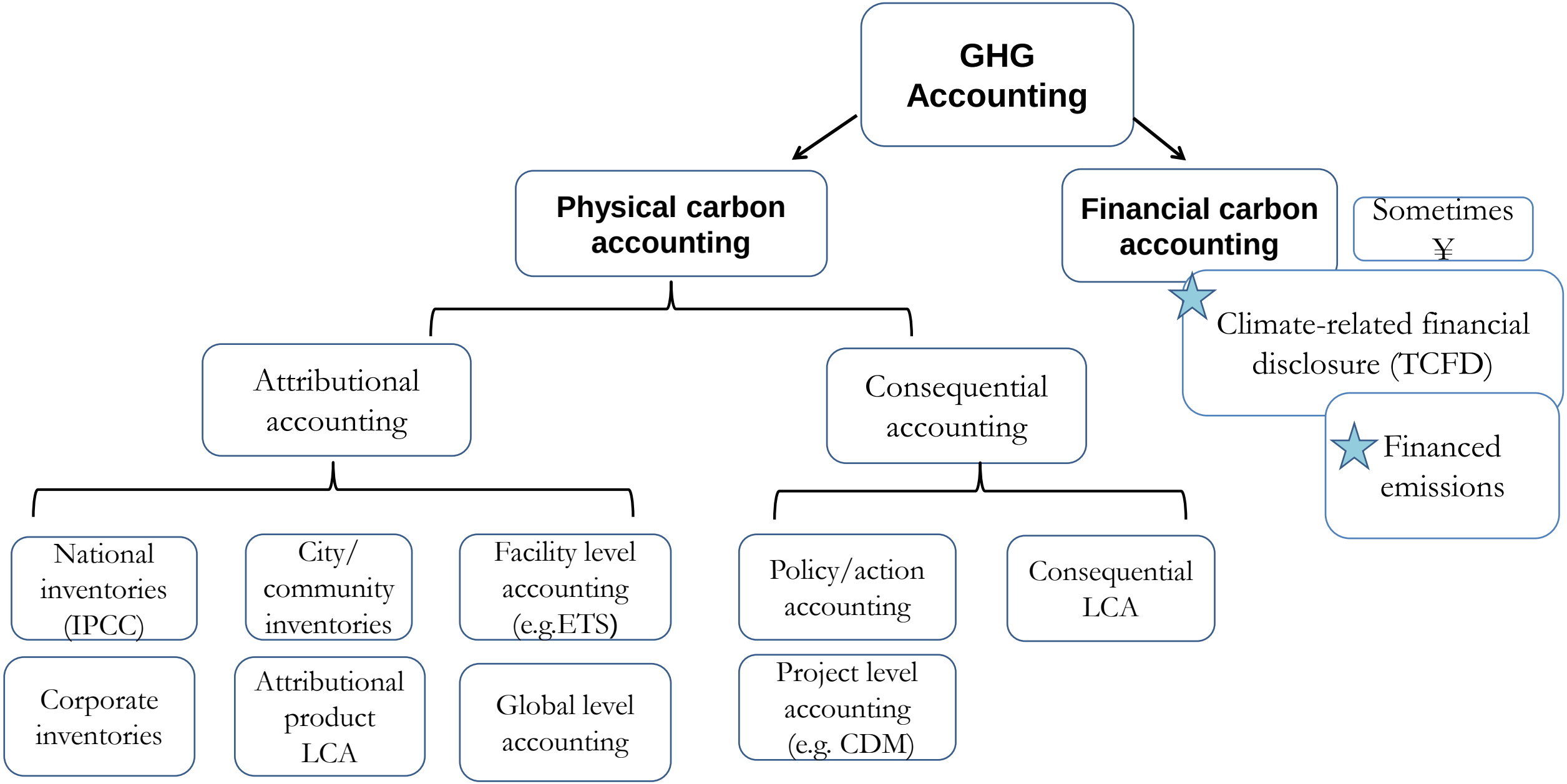
An auto parts manufacturer applies for a bank loan to expand its production.

The factory uses large amounts of electricity, and its major customers are asking suppliers to reduce GHG emissions.

The bank needs to assess whether the company can remain competitive and repay the loan over the next five years.

Question

How might the company's energy use and GHG emissions affect its future costs, sales, and ability to repay the loan??


GHG Accounting
Physical carbon accounting
Financial carbon accounting
Sometimes ¥
Climate-related financial disclosure (TCFD)
Financed emissions
Attributional accounting
Consequential accounting
National inventories (IPCC)
City/ community inventories
Facility level accounting (e.g.ETS)
Corporate inventories
Attributional product LCA
Global level accounting
Policy/action accounting
Consequential LCA
Project level accounting (e.g. CDM)

Overview of financial carbon accounting

- In a narrow sense, financial carbon accounting is the financial consequences imposed by emissions trading schemes. It involves with recognition (确认), measurement (计量) and disclosure (报告) issues on the treatment of carbon related assets and liabilities. (碳配额如何反应在财务三大表中)

- In a broad sense, financial carbon accounting is about climate-related financial disclosure, involving with identification and assessment of climate-related financial risks and opportunities.

- In our discussion, we will be focused on financial carbon accounting in the broad sense and look specifically on financed emissions.

目 录

Contents

01-Climate-Related Financial Risks

02-Methodology to Measure Financed Emissions

03-Example of financed emissions (*)

04-Wrap-up and Next Time

Learning Outcomes

1. **Illustrate** climate-related financial risks and opportunities with examples
2. **Explain** why financial institutions disclose climate-related information and account for emissions.
3. **Interpret** the basic logic of financed emissions accounting, including emission attribution, data requirements, and limitations。

01

Climate-Related Financial Risks

Physical impacts of climate change

- World Meteorological Organization (2025). WMO Statement on the State of the Global Climate 2025. [online] Geneva: WMO. Available at: https://wmo.int/publication-series/state-of-global-climate/state-of-global-climate-2025
  - 2015-2025 are the hottest 11-years on record
  - Earth's energy imbalance is highest in sixty five-year record
  - The ocean has been absorbing about eighteen times the annual human energy use each year for the past two decades
  - Annual sea ice extent in the Arctic was at or near a record low
  - Extreme events around the world, including intense heat, heavy rainfall and tropical cyclones, caused disruption and devastation and highlighted the vulnerability of our inter-connected economies and societies.
- Predicting climate impacts:
  - Scientists can create complex climate models to help predict what may happen in the future, e.g., Coupled Model Intercomparison Project (CMIP)
  - Businesses are also looking at various possible climate change scenarios and their impacts, e.g., Shell https://www.shell.com/energy-and-innovation/the-energy-future/scenarios.html



Shell Scenarios

Is hydrogen the future? What is the role for biofuels? How long until all the world's cars are electric? We explore how the future could unfold to help people make better decisions today. By offering different possible versions of the future, our work stretches minds, confronts assumptions and helps people think differently about the world around them. Our scenarios are not predictions about the future, do not involve a crystal ball and are not Shell strategy. Even so, the leadership teams in Shell have been using this work to help make decisions for more than 50 years.

Business risks from climate change

- Assets can be directly damaged by floods, droughts and severe storms, but portfolios can also be harmed indirectly, through weaker growth and lower asset returns.
  - the *ClimateVaR* estimated by *the Economist* (2015) illustrated that the value at risk, as a result of climate change, to the total global stock of manageable assets as ranging from $4.2 trillion to $43 trillion between now and the end of the century
  - Stranded assets: “unburnable carbon”
    - if climate change is to be limited, the substantial portion of known coal, oil and gas reserves will probably remain unburned

A clear understanding of climate-related risks is essential for the financial sector.

Climate-related risks

- Climate related risk encompasses several different types of risk
  - Transition risk (转型风险)
    - E.g., changes in law, policy, technology and market related to the transition to a low-carbon economy
  - Physical risk (物理风险)
    - E.g., damages to fixed assets, supply chain disruption

Table 1: Primary Types of Carbon Risk Factors

Category of Risk	Definition	Nature of Impact	Examples
Policy and Legal	Policies or regulations that could impact the operational and financial viability of carbon assets	Impacts physical carbon assets and companies that own/ operate assets	Fuel-efficiency standards for personal vehicles; emissions trading systems; U.S. EPA regulations targeting air pollution and GHGs from power plants
Technology	Developments in the commercial availability and cost of alternative and low-carbon technologies	Impacts technology choices, deployment and costs and demand profiles	Energy storage technologies; advances in renewable energy technologies, carbon capture and storage; alternative fuels
Market and Economic	Changes in market or economic conditions that would negatively impact carbon assets	Impacts physical carbon assets and companies that own/ operate assets	Changes in fossil fuel prices; changes in consumer preferences

WRI and UNEP-FI Portfolio Carbon Initiative. Carbon Asset Risk: Discussion Framework. [online] Washington, DC: WRI and Geneva: UNEP-FI. Available at: http://www.unepfi.org/fileadmin/documents/carbon_asset_risk.pdf

Climate-related opportunities

- Small-scale examples: recyclable coffee cups, English wine…
- Large-scale examples:
  - HSBC: sustainable finance (facilitate 100 billion of new capital in sustainable finance by 2025; reduce exposure to thermal coal and are moving to 100% renewables from an operational perspective by 2030)
  - The business case for clean energy
  - Renewable energy in developing economies
  - The emerging green economy

Climate risk and opportunities - Activity

Organize the risks and opportunities into the correct categories.

Transition Risk	Physical Risk	Opportunities
(1) (6) (7)	**(2) (4)**	**(3) (5)**

(1) Introduction of a carbon price leads to increased operating costs.

(2) Rising sea levels inundate operations located in coastal regions.

(3) Introduction of government subsidies for companies that have “green” portfolios.

(4) More frequent and severe storms damage facilities and infrastructure.

(5) Considerations of climate change-related risks and opportunities throughout the organization builds climate resilience.

(6) Reputational damages due to stigmatization of sector from negative stakeholder feedback.

(7) New regulatory requirements addressing climate change increase compliance requirements.

TCFD: Task Force on Climate-related Financial Disclosures

- In 2015, Mark Carney, in his position as chair of the FSB, established an industry-led task force - the TCFD - to identify the information needed by investors, lenders and insurances underwriters to assess and price climate-related risks and opportunities.

- The TCFD was asked to develop voluntary, consistent climate-related financial disclosures that would be useful in understanding material risks related to climate change.

- In 2017, the TCFD released its final report on recommendations which included **four core elements** with **eleven recommended disclosures**.

- The TCFD formally fulfilled its remit and disbanded in October 2023, with its monitoring role transferred to the IFRS Foundation from 2024 and its core recommendations integrated into the ISSB/IFRS sustainability disclosure framework.



Core elements of disclosures from the TCFD recommendations

Governance

Governance around climate-related risks and opportunities

1. How the board provides oversight
2. How the organisation's management assesses and manages them

Strategy

Actual and potential impacts of climate-related risks and opportunities on business strategy and financial planning

1. What risks and opportunities have been identified over the short, medium and long-terms
2. The impact on the organisation's business, strategy and financial planning
3. How resilient the strategy is under different climate scenarios

Risk management

The processes used to identify, access and manage climate related risks

1. Identifying and assessing climate-related risks
2. Managing them
3. Integrating climate risk management into

Metrics and targets

The metrics and targets used to assess and manage relevant climate-related risks and opportunities

1. Scope 1 and Scope 2 greenhouse gas (GHG) emissions (Scope 3 emissions should be disclosed 'if appropriate')
2. Climate-related targets and process made in achieving them

Figure 1: Core elements of TCFD disclosures

Financial sector group
Banks
Insurers
Asset Owners
Asset Managers



Financed Emissions

Emissions generated as a result of financial services, investments, and lending by investors and companies that provide financial services.

Non-financial sector group	
Energy	– Oil and Gas – Coal – Electric Utilities
Transportation	– Air Freight – Passenger Air Transportation – Maritime Transportation – Rail Transportation – Trucking Services – Automobiles and Components
Materials and Buildings	– Metals and Mining – Chemicals – Construction Materials – Capital Goods – Real Estate Management and Development
Agriculture, Food and Forest Products	– Beverages – Agriculture – Packaged Foods and Meats – Paper and Forest Products

02

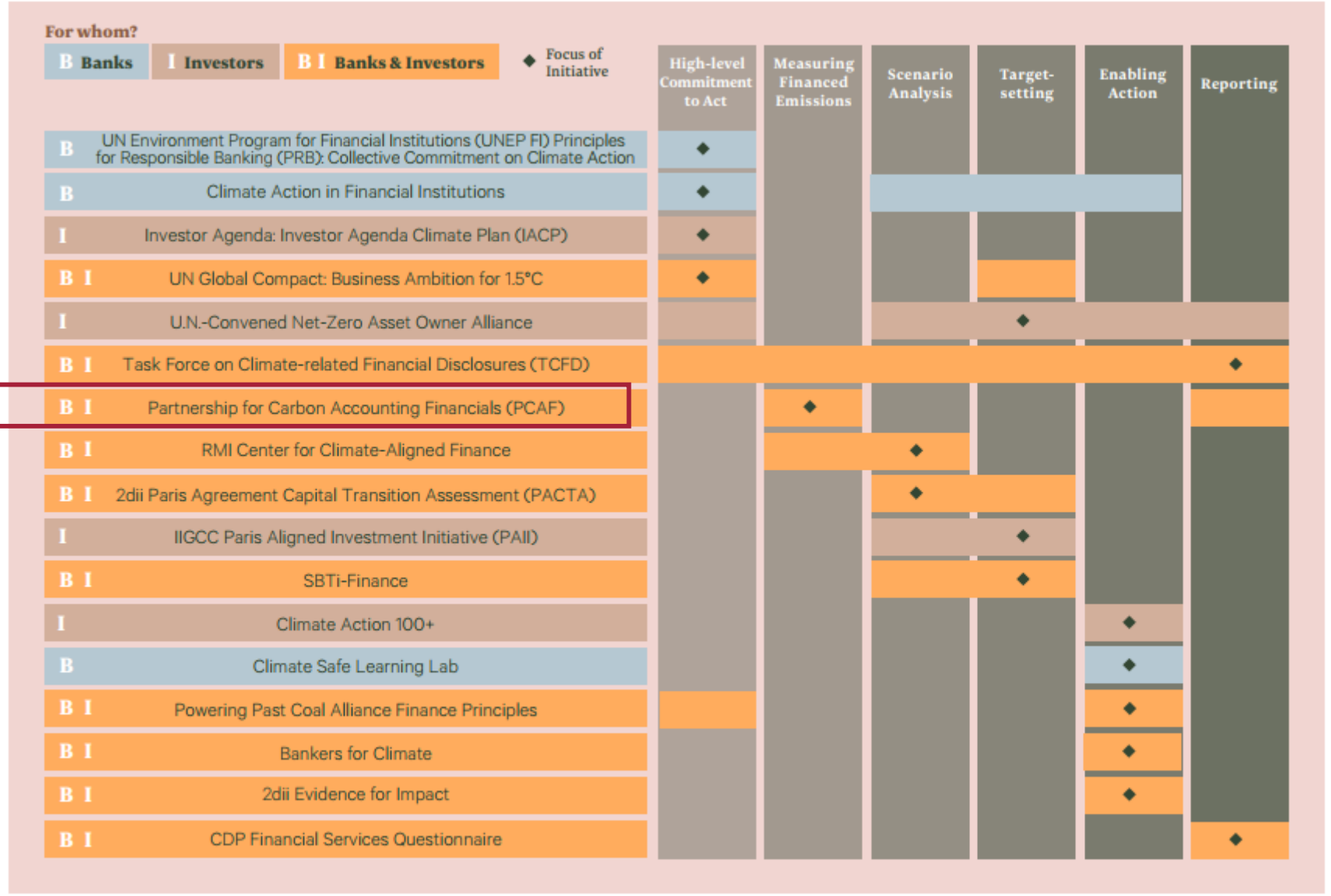
Methodology to Measure Financed Emissions

Challenges in measuring financed emissions

- Challenges:
  - Financial institutions are not familiar with carbon accounting.
  - It is a burdensome and complex process.
  - Limited data.
  - Scrutiny over the additionality and fungibility of green bonds
  - Lack of standardized legislation, policies, methodologies, and tools when measuring impact

- Until recently, there was no widely accepted global standard for measuring and disclosing financed emissions comparable to the GHG Protocol or ISO standards for corporate GHG emissions accounting.

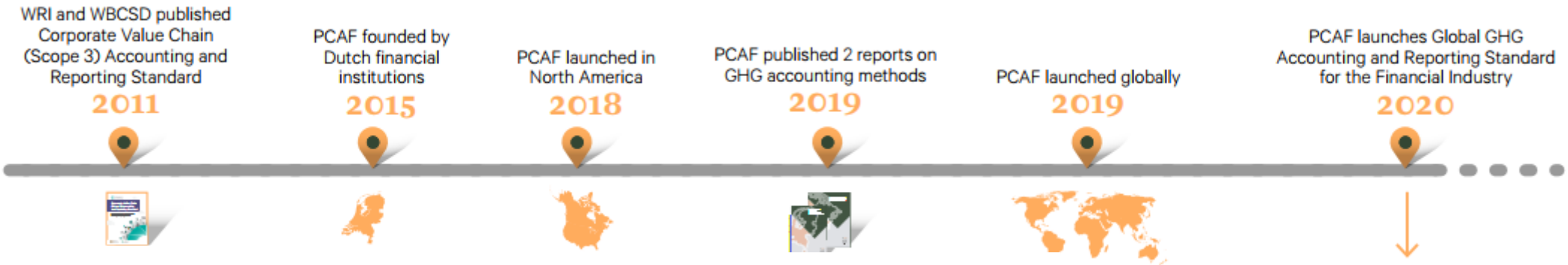
Cluster of climate initiatives for financial sectors



Source: (PCAF, 2020)

Background of PCAF

- The Partnership for Carbon Accounting Financials (PCAF) (碳核算金融联盟) bring financial institutions together to better measure and address carbon emissions.





Over 85 financial institutions, with over $ 17 Trillion in financial assets committed (November 2020).

BUILT ON GHG PROTOCOL

This Standard was reviewed by the GHG Protocol and is in conformance with the requirements set forth in the Corporate Value Chain (Scope 3) Accounting and Reporting Standard, for Category 15 investment activities.

There are nearly 300 PCAF members who've committed to measuring and disclosing their financed emissions.

JOIN PCAF

EXPLORE PCAF'S STANDARD

The GLOBAL GHG ACCOUNTING & REPORTING Standard



A
Financed Emissions

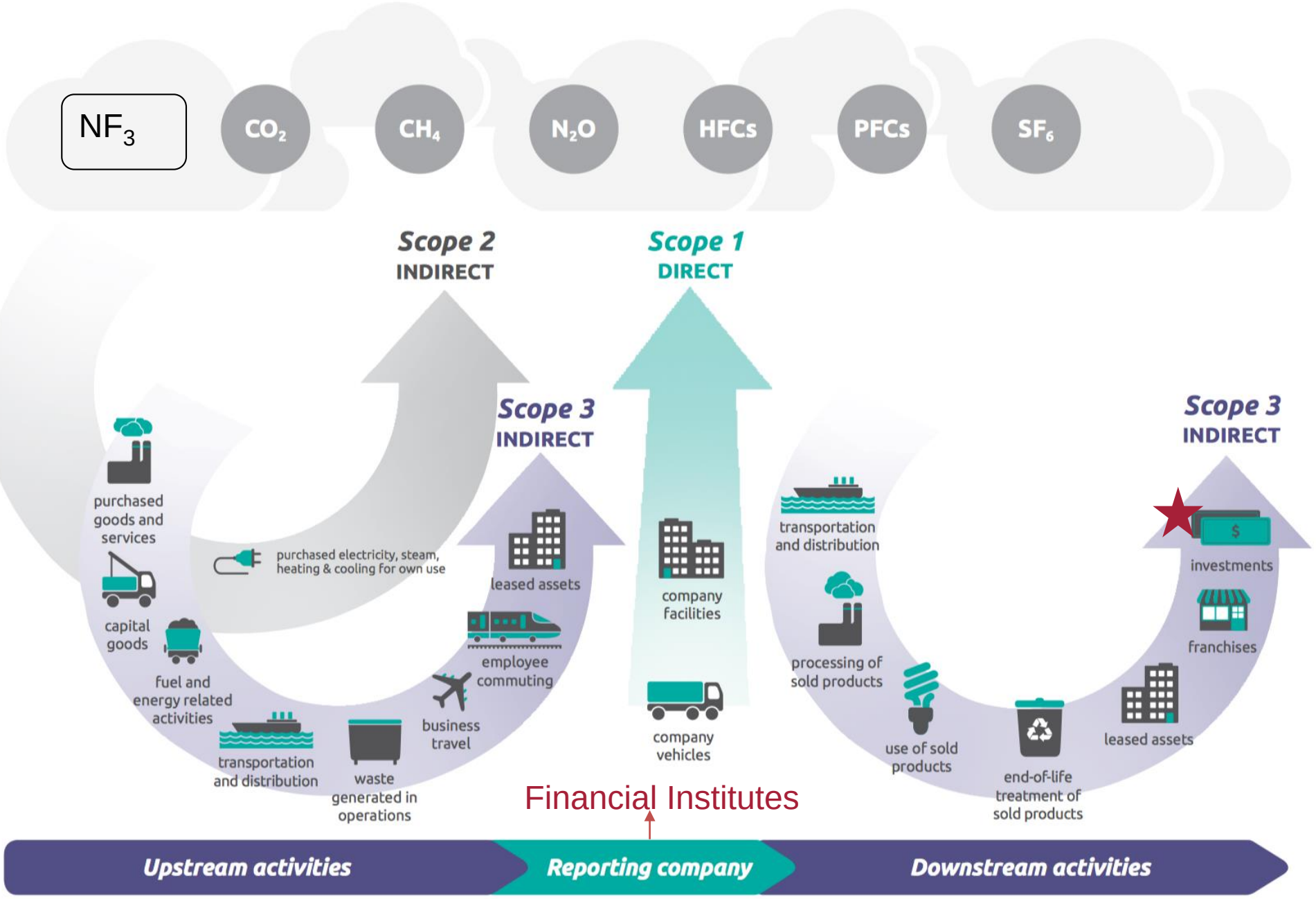
B
Facilitated Emissions

C
Insurance-Associated Emissions

700 + financial institutions take action and share knowledge.

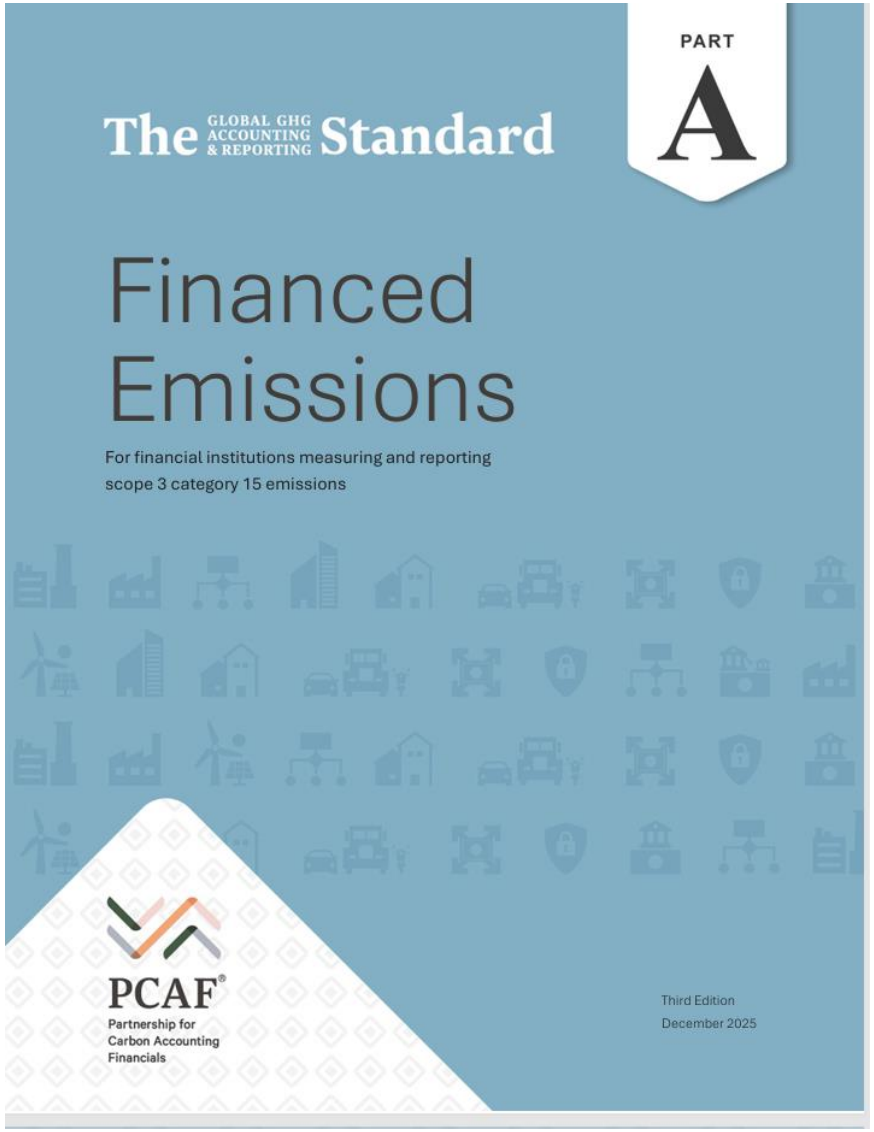
Join the global initiative to strengthen transparency and accountability in the financial sector.

Figure [1.1] Overview of GHG Protocol scopes and emissions across the value chain



Quantify financed emissions

- The Global GHG Accounting and Reporting Standard for the Financial Industry was developed by the PCAF in 2020 and updated twice.
  - The goal is to create and implement a streamlined approach to disclosing GHG emissions resulting from investments and loans.
  - Providing formulas for allocating the carbon impacts for transactions in six major asset classes. This ensures that financed emissions calculations are comparable across the whole finance industry.
- Apart from financed emissions, financial institutions shall also measure and report their scope 1 and 2 emissions.



3rd version (2025)

Principles for measurement of financed emissions

GHG Protocol principles for Scope 3 inventories

- Completeness
- Consistency
- Relevance
- Accuracy
- Transparency

Additional PCAF requirements

- Recognition (识别，即方法的选择)
- Measurement （衡量）
- Attribution（归因）
- Data quality（数据质量）
- Disclosure（披露）

Double counting in financed emissions

- occurs when GHG emissions are counted more than once in the financed emissions calculation of one or more institutions

- usually take place at following five levels:
  - Between financial institutions
  - In cofinancing the same entity or activity
  - Between transactions within the same financial institutions
  - Across different asset classes
  - Within the same asset class

- cannot be avoided, but should be minimized as much as possible

Methodologies for measurement of financed emissions (*)

- The PCAF defines financed emissions as the annual corporate accounting and disclosure of GHG emissions financed by loans and investments in the portfolio of a financial institution at a fixed point in time in line with financial accounting periods.

- The standard by PCAF provides detailed methodological guidance for ten asset classes. Although different assets involve with different financial products and detailed method, they share the key idea and core methodology.

上市证券与债券

Listed Equity and Corporate Bonds

$$\frac{\text{Outstanding amount}}{\text{EVIC or Total company equity + debt}} \times \text{Company emissions}$$

EVIC = enterprise value including cash

商业贷款

Business Loans and Unlisted Equity

$$\frac{\text{Outstanding amount}}{\text{EVIC or Total company equity + debt}} \times \text{Company emissions}$$

EVIC = enterprise value including cash

项目融资

Project Finance

$$\frac{\text{Outstanding amount}}{\text{Total project equity + debt}} \times \text{Project emissions}$$

GHG accounting for six asset classes

商业地产贷款

Commercial Real Estate

$$\frac{\text{Outstanding amount}}{\text{Property value at origination}} \times \text{Building emissions}$$

房贷

Mortgages

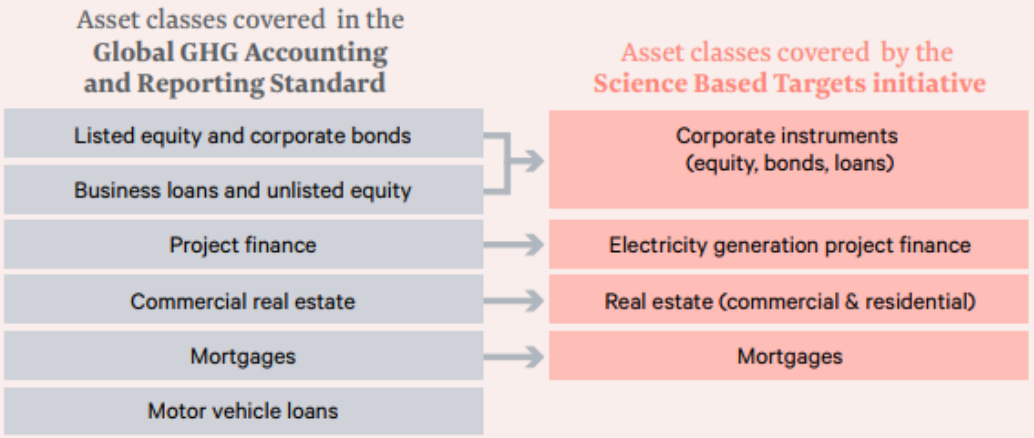
$$\frac{\text{Outstanding amount}}{\text{Property value at origination}} \times \text{Building emissions}$$

车辆贷款

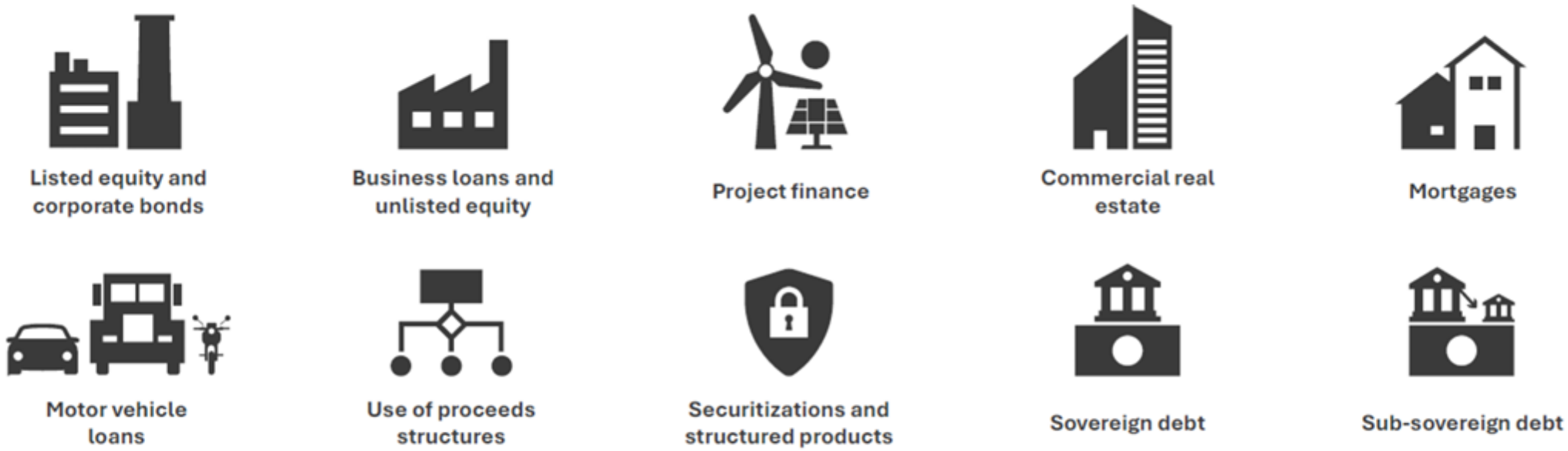
Motor Vehicle Loans

$$\frac{\text{Outstanding amount}}{\text{Total value at origination}} \times \text{Vehicle emissions}$$

Figure 3-4. Asset classes covered by PCAF and SBTi



PCAF (2020)

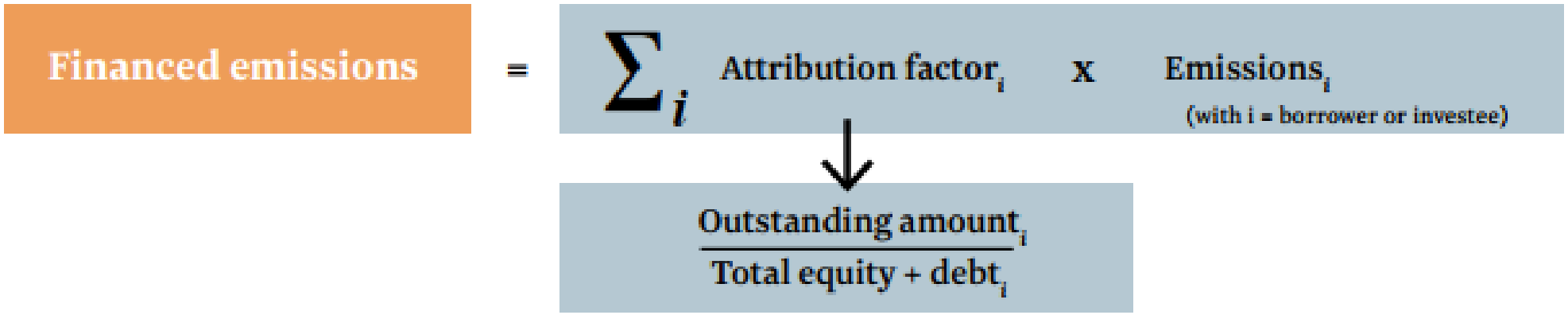


Each asset class has its own section covering methodological guidance on the following elements:[32]

- Asset class definition
- Emission scopes covered
- Attribution of emissions
- Equations to calculate financed emissions
- Data required
- Other considerations
- Limitations

PCAF (2025)

Figure 4-2. The general approach to calculate financed emissions



03

Example of financed emissions

Example of calculation for financed emissions

张佳亮和魏天歌（2021）金融机构碳核算发展展望. 金融纵横. 2021, 05.

- Bank A (investor) and Company B (investee) in China

Bank A provided 5 000 thousand CNY business loans to Company B in 2020.

The total assets of Company B in 2020 (Assets = Liabilities + Equity) were 50 000 thousand CNY. The total emissions reported from Company B were 100 thousand tons.

- According to PCAF, Bank A's financed emissions from Company B can be calculated by

$$\text{银行A 融资排放} = \frac{\text{500万元（银行A提供给企业B的总贷款）}}{\text{5000万元（企业B的总企业价值）}} \times \text{10万吨（企业B的总年排放）}$$

归因因子=0.1

图 3　银行 A 对企业 B 融资的排放量

Bank A (investor) and Company B (investee)

In the same example, if Company B did not report emissions data, but can provide data on energy consumption such that 100 000 MWh grid electricity and 100 TJ natural gas consumed in 2020.

Then Bank A can go with Option 2 to calculated physical activity-based emissions. Assume that Bank A collected the emission factor of natural gas in Asia from PCAF database, and the grid emission factor from relevant authorities. Then the financed emissions can be calculated by

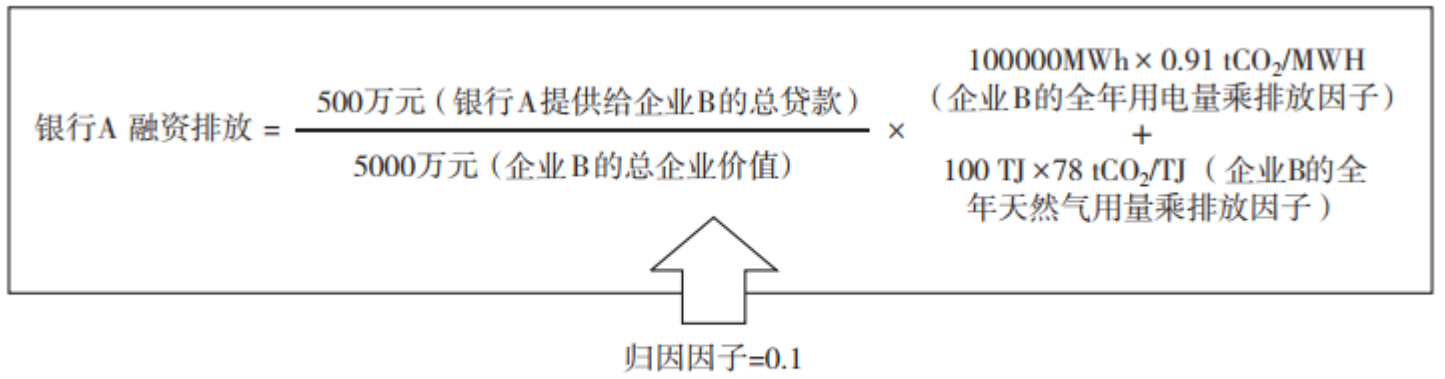


图 5　利用选项 2 测算的银行融资排放

04

Wrap Up and Next Time

Wrap up

Climate-related risks

- Physical risk
- Transition risk

Climate-related financial disclosure:

- TCFD → ISSB/IFRS sustainability disclosure framework

Financed emissions measurement:

- PCAF methods





See you next time